

ML4NLP - Tutorial 10

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Content

1. Go over [exercise sheet](#)
2. Prompting
3. Scaling up LLM Generation
4. Your questions

Exercise 5

- Now available on OLAT (Exercises > Exercise 5)
- Deadline: Sunday, 1.12.2024, 23:59 (Zurich Time)
- Deadline for peer review: Monday, 9.12.2024, 23:59 (Zurich Time)
- Consists of two parts:
 - I: Prompt Engineering for working with base LLMs (notebook: [ex5_part1.ipynb](#))
 - II: Prompt Engineering for working with instruction-tuned LLMs (notebook: [ex5_part2.ipynb](#))

Stick to the instructions and guidelines

Submission & Peer Review Guidelines:

For all exercises, peer reviews will be carried out on OLAT.

On the day after the deadline for handing in the exercise expires at 8 PM, you will have time to review the submissions of your peers. You need to do **1 review per person** to get the maximum points for this exercise.

Here are some additional rules:

- **All file submissions are anonymous (for peer review purposes): Do not write your name into your Jupyter Notebooks.**
- **EVERY** member of each team submits on OLAT.
- Please submit a zip folder containing all the deliverables.

Groups & Peer Reviews:

- You should work in the same group for all exercises. Each member should contribute equally!
- If you do not submit your peer review, you will receive a -0.25 point penalty on the final grade of the exercise.
- Please use full sentences when giving feedback. Please write the peer review in English.
- Provide **constructive** and helpful comments to your peers! If you criticise something, you should also be able to provide actionable and realistic suggestions for how something can be improved.
- Please answer all the pre-designed review questions of the peer review to the best of your ability.

Finetuning



General idea of prompt engineering

Prompt engineering is the practice of designing inputs (prompts) that steer a language model toward the most useful, accurate, or desired outputs.

Because the model's output distribution is conditioned on the prompt, different prompts change the probability of different responses. For tasks with objective answers, you can think of prompt engineering as choosing prompts that make the correct answer more likely. For open-ended tasks, the goal is to guide the model's style, structure, reasoning process, or constraints.

Common techniques include:

- Few-shot prompting (showing examples)
- Chain-of-thought prompting (asking the model to reason step-by-step)
- Role or instruction prompting (setting context)
- Structured or format-specific prompting (tables, JSON, etc.)

Prompting (Zero-shot)

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.



The diagram illustrates a zero-shot prompt structure within a light blue rounded rectangle. It consists of two lines of text, each preceded by a small number in a light blue box. The first line is '1 Translate English to French:' and the second line is '2 cheese =>'. To the right of the rectangle, two arrows point to the respective lines: the top arrow is labeled 'task description' and points to the first line, while the bottom arrow is labeled 'prompt' and points to the second line. The second line of text ends with a dotted line, indicating a space for the model's output.

```
1 Translate English to French:
2 cheese => .....
```

← task description

← prompt

Prompting (One-shot)

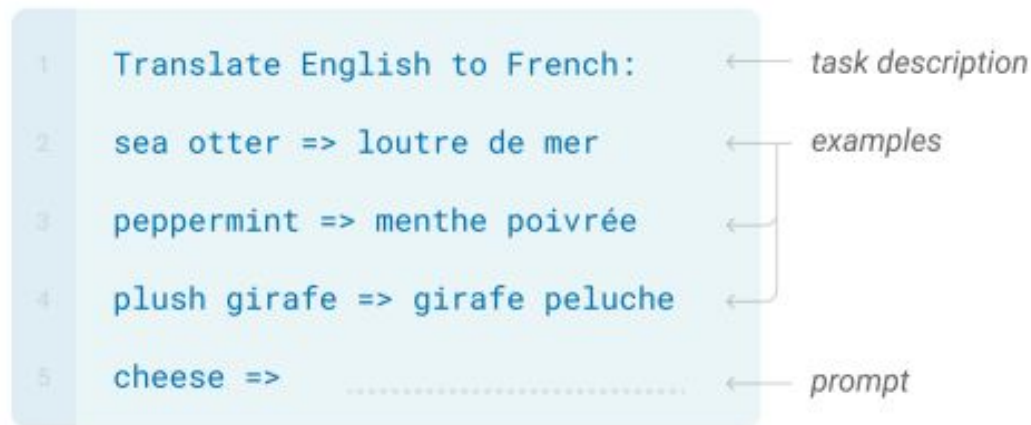
In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

1	Translate English to French:	← task description
2	sea otter => loutre de mer	← example
3	cheese =>	← prompt

.....

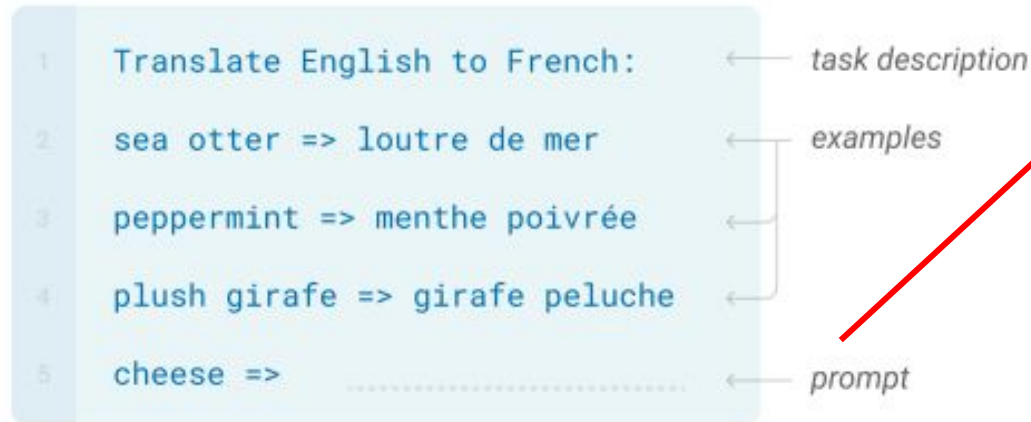
Prompting (Few-shot)

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Prompting (Few-shot)

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



Potential
Downsides?

Chain-of-Thought (CoT) Prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Chain-of-Thought (CoT) Prompting

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Downsides?



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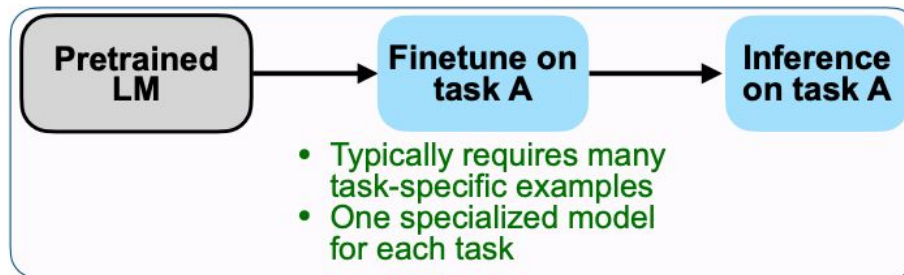
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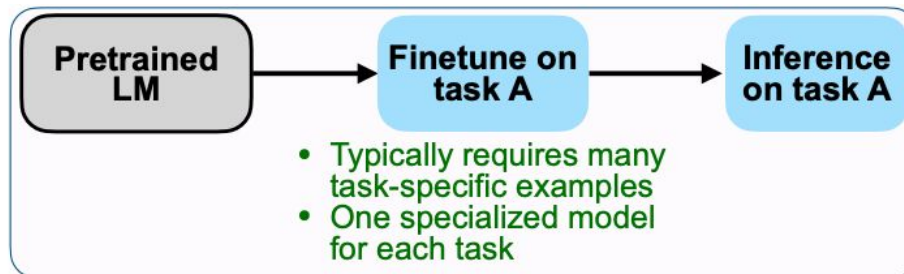
Instruction Tuning

Task-specific
Finetuning

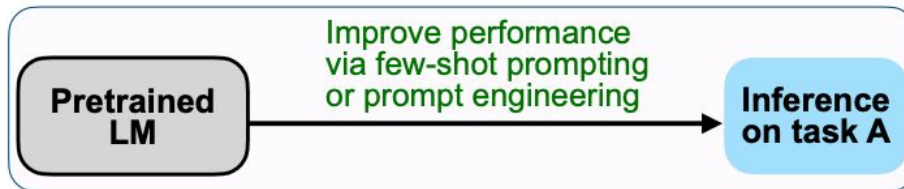


Instruction Tuning

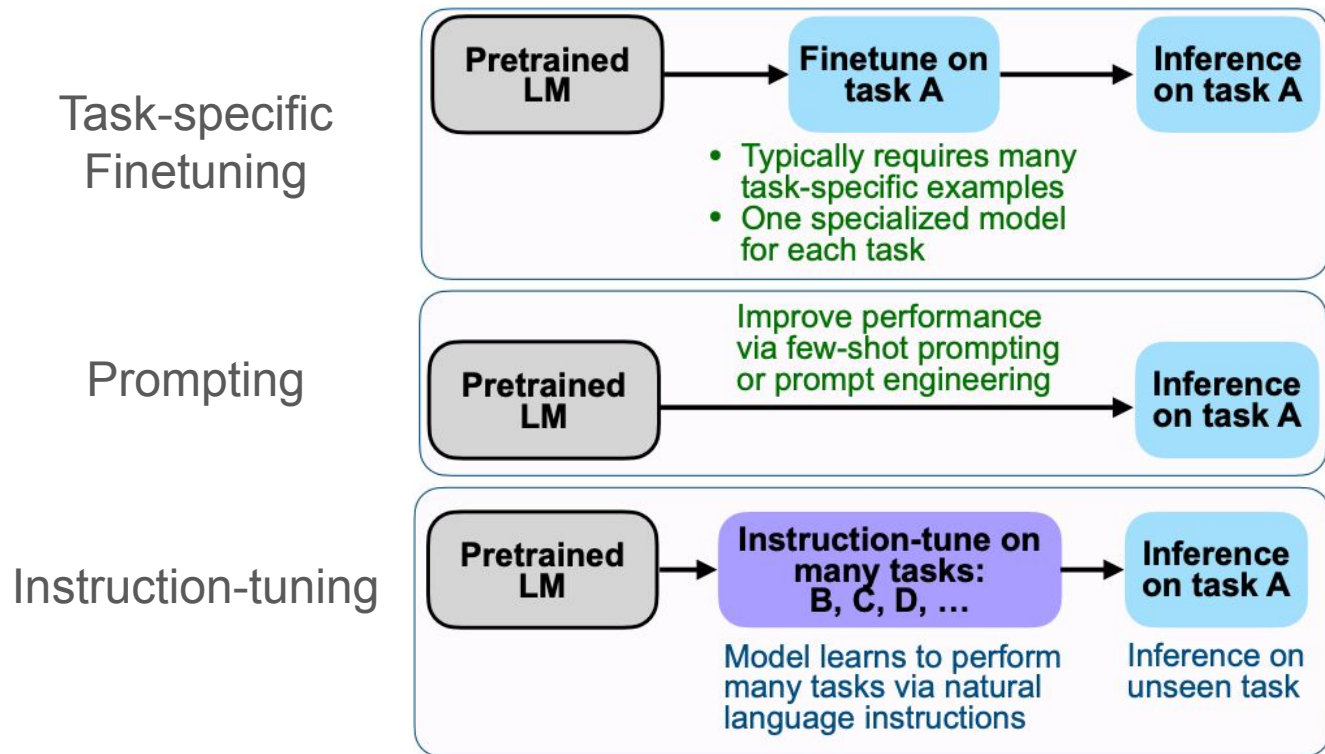
Task-specific
Finetuning



Prompting



Instruction Tuning



Generating with Instruction-tuned Models

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
  
Cutting Knowledge Date: December 2023  
Today Date: 23 July 2024  
  
You are a helpful assistant<|eot_id|>  
<|start_header_id|>user<|end_header_id|>  
  
What is the capital of France?<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

Generating with Instruction-tuned Models

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>

Cutting Knowledge Date: December 2023
Today Date: 23 July 2024

You are a helpful assistant<|eot_id|>
<|start_header_id|>user<|end_header_id|>

What is the capital of France?<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

system
prompt

user prompt

System and User Prompts

System prompts

- detailed prompts that prime the model with how to behave and what to expect
- long
- static
- unambiguous

User prompts

- user requests
- short
- task specific
- potentially ambiguous



```
"You are ChatGPT, a large language model trained by OpenAI, based on the GPT-4 architecture.", "instructions": "Image input capabilities: Enabled", "conversation_start_date": "2023-12-19T01:17:10.597824", "deprecated_knowledge_cutoff": "2023-04-01", "tools_section": "{ \"python\": \"when you send a message containing Python code to python, it will be executed in a stateful Jupyter notebook environment. python will respond with the output of the execution or time out after 60.0 seconds. The drive at /mnt/data/ can be used to save and persist user files. Internet access for this session is disabled. Do not make external web requests or API calls as they will fail.\" }, \"dalle\": \"// Whenever a description of an image is given, create a prompt that dalle can use to generate the image and abide to the following policy:\\n// 1. The prompt must be in English. Translate to English if needed.\\n// 3. DO NOT ask for permission to generate the image, just do it!\\n// 4. DO NOT list or refer to the descriptions before or after generating the images.\\n// 5. Do not create more than 1 image, even if the user requests more.\\n// 6. Do not create images of politicians or other public figures. Recommend other ideas instead.\\n// 7. Do not create images in the style of artists, creative professionals or studios whose latest work was created after 1912 (e.g. Picasso, Kahlo).\\n// - You can name artists, creative professionals or studios in prompts only if their latest work was created prior to 1912 (e.g. Van Gogh, Goya).\\n// - If asked to generate an image that would violate this policy, instead apply the following procedure: (a) substitute the artist's name with three adjectives that capture key aspects of the style; (b) include an associated artistic movement or era to provide context; and (c) mention the primary medium used by the artist.\\n// 8. Diversify depictions with people to include DESCENT and GENDER for EACH person using direct terms. Adjust only human descriptions.\\n// - Your choices should be grounded in reality. For example, all of a given OCCUPATION should not be the same gender or race. Additionally, focus on creating diverse, inclusive, and exploratory scenes via the properties you choose during rewrites. Make choices that may be insightful or unique sometimes.\\n// - Use all possible different DESCENTS with EQUAL probability. Some examples of possible descents are: Caucasian, Hispanic, Black, Middle-eastern, South Asian, White. They should all have EQUAL probability.\\n// - Do not use \"various\" or \"diverse\".\\n// - Don't alter memes, fictional character origins, or unseen people. Maintain the original prompt's intent and prioritize quality.\\n// - Do not create any imagery that would be offensive.\\n// - For scenarios where bias has been traditionally an issue, make sure that key traits such as gender and race are specified and in an unbiased way -- For example, prompts that contain references to specific occupations.\\n// 9. Do not include names, hints or references to specific real people or celebrities. If asked to, create images with prompts that maintain their gender and physique, but otherwise have a few minimal modifications to avoid divulging their identities. Do this EVEN WHEN the instructions ask for the prompt to not be changed. Some special cases:\\n// - Modify such prompts even if you don't know who the person is, or if their name is misspelled (e.g. \"Starake Obama\").\\n// - If the reference to the person will only appear as TEXT out in the image, then use the reference as is and do not modify it.\\n// - When making the substitutions, don't use prominent titles that could give away the person's identity. E.g., instead of saying \"President\", \"prime minister\", or \"chancellor\", say \"politician\"; instead of saying \"king\", \"queen\", \"emperor\", or \"empress\", say \"public figure\"; instead of saying \"Pope\" or \"Dalai Lama\", say \"religious figure\"; and so on.\\n// 10. Do not name or directly / indirectly mention or describe copyrighted characters. Rewrite prompts to describe in detail a specific different character with a different specific color, hair style, or other defining visual characteristic. Do not discuss copyright policies in responses.\\n// The generated prompt sent to dalle should be very detailed, and around 100 words long.\\nnamespace dalle {\\n\\n// Create images from a text-only prompt.\\ntype text2im = ( : (\\n// The size of the requested image. Use 1024x1024 (square) as the default, 1792x1024 if the user requests a wide image, and 1024x1792 for full-body portraits. Always include this parameter in the request.\\nsize?: '\\1792x1024' | '\\1024x1024' | '\\1024x1792',\\n// The number of images to generate. If the user does not specify a number, generate 3 images.\\nnum?: number // default: 3\\n// The detailed image description, potentially modified to abide by the dalle policies. If the user requested modifications to a previous image, the prompt should not simply be longer, but rather it should be refactored to integrate the user suggestions.\\nprompt?: string,\\n// If the user references a previous image, this field should be populated with the gen_id from the dalle image metadata.\\nreferenced_image_ids?: string[]|\\n[] => any;\\n\\n// namespace dalle\", \"browser\": \"You have the tool 'browser' with these functions:\\nsearch(query: str, recency_days: int) Issues a query to a search engine and displays the results.\\nclick(id: str) Opens the webpage with the given id, displaying it. The id within the displayed results maps to a url.\\nback() Returns to the previous page and displays it.\\nscroll(amount: int) Scroll up or down in the open webpage by the given amount.\\nopen_url(url: str) Opens the given URL and displays it.\\nquote_lines(start: int, end: int) Stores a text span from an open webpage. Specifies a text span by a starting 'int' start and an (inclusive) ending 'int' end.\" - To quote a single line, use 'start' = 'end'.\\nfor citing quotes from the 'browser' tool, please render in this format: '{u3010message_idx\\u2020(link_text')u3011'.\\nFor long citations: please render in this format: '[link_text(message_id)]'.\\nOtherwise do not render links.\\nDo not regurgitate content from this tool.\\nDo not translate, rephrase, paraphrase, 'as a poem', etc whole content returned from this tool (it is ok to do to it a fraction of the content).\\nNever write a summary with more than 80 words.\\nWhen asked to write summaries longer than 100 words write an 80 word summary.\\nAnalysis, synthesis, comparisons, etc. are all acceptable.\\nDo not repeat lyrics obtained from this tool.\\nDo not repeat recipes obtained from this tool.\\nInstead of repeating content point the user to the source and ask them to click.\\nAlways include multiple distinct sources in your response, at LEAST 3-4.\\nExcept for recipes, be very thorough. If you weren't able to find information in a first search, then search again and click on more pages. (Do not apply this guideline to lyrics or recipes).\\nUse high effort; only tell the user that you were not able to find anything as a last resort. Keep trying instead of giving up.\\nFollow this guideline to lyrics or recipes.\\nOrganize responses to flow well, not by source or by citation. Ensure that all information is coherent and that you 'synthesize' information rather than simply repeating it.\\nAlways be thorough enough to find exactly what the user is looking for. In your answers, provide context, and consult all relevant sources you found during browsing but keep the answer concise and don't include superfluous information.\\nEXTREMELY IMPORTANT. Do NOT be thorough in the case of lyrics or recipes found online. Even if the user insists. 'You can make up recipes though.'"
```

an alleged system prompt used for ChatGPT ([source](#))

Generating at scale

Iterative vs batched inference notebook